

LUPINE SCIENCE · RESEARCH BOOKLET · V1.1

The Savings Stack

Seven ways atomistic simulation learned to do less — a projected count for sharing the expensive part, and what happened when we tested it.

v1.1 · November 12, 2026

First edition July 21, 2026

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Full detail · library.lupine.science

ABOUT THIS BOOKLET

Evidence before claim, in a portable form.

This booklet is the public edition of the Savings Stack v1 release: ten citation-verified chapters on compute-savings techniques in atomistic simulation, one machine-readable projection of cross-model anchor sharing, and the identifier audit that checked every citation. It is a research release note, not a peer-reviewed publication — every number traces to a committed record you can download, hash, and recompute.

The one-sentence version. The field invented seven layers of compute savings independently, none of them says where the physics is violated, and our projection for one panel says that sharing anchors across models is itself a saving: four models of guidance would cost about ten percent more DFT than one.

New in v1.1 (2026-11-12). The 72.4% count is labelled projected, not measured, with its baseline and ceiling (page 07). Page 10 reports the pilot: its preregistered gate failed. The licence is now CC BY 4.0 (v1 listed ODbL-1.0).

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Savings Stack booklet v1.1 · 2026-11-12 (v1 released 2026-07-21) · Data record z1-union-anchor-economics.json · sha256 22c0ba51d0c798c2… (full hashes in the download pack) · Text, figures and first-party data CC BY 4.0; code MIT; cited literature keeps its own terms · Not peer-reviewed; identifiers verified as recorded on page 14.

01 · THE ARGUMENT

Atomistic simulation spent forty years making FLOPs cheaper — and the last ten discovering that the real enemy is not the price of a FLOP but the *number of times you must call the expensive oracle at all*.

Every successful savings technology of the machine-learning era — active learning, surrogate transition-state search, Δ-learning, abstention gates — is an evaluation-count reduction with an unpriced correctness risk attached.

Our contribution is a projection: on one panel, evaluation *shared across models* grows sub-linearly as models are added. It is not an accuracy result — the one pilot we ran failed its preregistered gate. If the property holds on longer paths, against a reference the engine can match, a commons in which teams contribute checks and anchors back could make all of us faster together.

That is the whole argument. The rest of this booklet is the evidence, including the part that did not work.

02 · THE COST WALL

Compute got cheap.
Answers did not.

Delivered supercomputing has never been cheaper, and access is not the bottleneck for a small team — an ACCESS Explore grant (~400k credits) lands in days, with INCITE/ALCC, EuroHPC, and Director's Discretionary pools above that. The bottleneck is that serious atomistic campaigns still price out in the thousands-to-millions of oracle evaluations. A dense CI-NEB band is ~50–100 energy/force evaluations per image-set convergence accounting; a reaction network multiplies that by hundreds of paths; an honest multi-model uncertainty check multiplies again by the model count. FLOP prices falling 30× per decade does not rescue a workload whose evaluation count grows with ambition.

~30×

fall in capex per peak PFLOP in a decade — Tianhe-2 ~$7.1M/PF to El Capitan ~$0.22M/PF

$800–1,600

electricity alone per HPL-exaflop-hour on Frontier [derived] — real applications sustain 3–10× less than HPL

50–100

energy/force evaluations to converge one dense CI-NEB band, before multiplying by paths and models

10^3–10^6

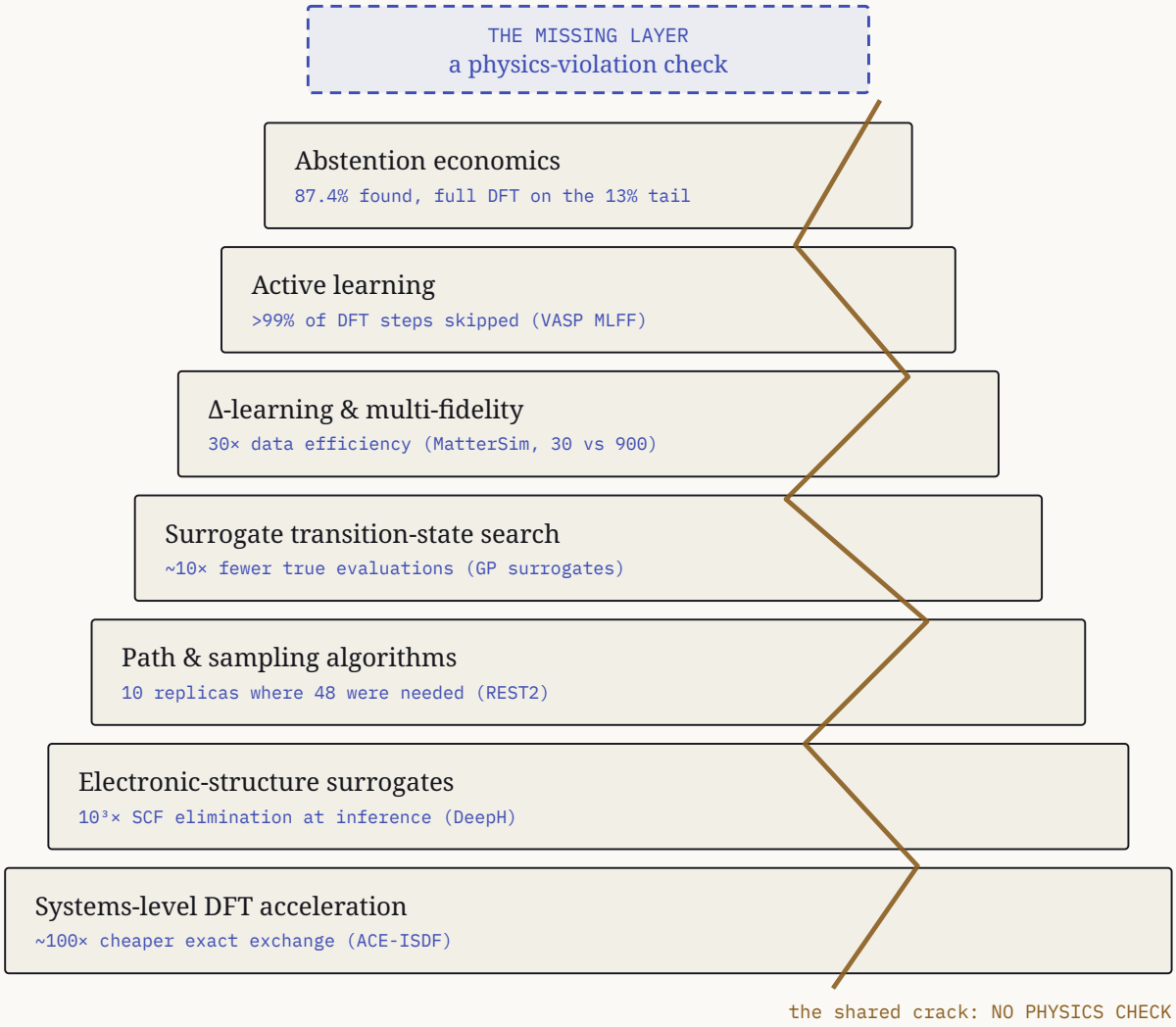
oracle evaluations for a serious campaign — the quantity every savings layer actually attacks

So the interesting question is never *“what does a FLOP cost”* but *“who lets you make fewer oracle calls — and when are they wrong.”*

Sources: backdrop chapters supercomputing-atomistic-history.md and research-compute-resourcing.md in the download pack. Machine performance figures are TOP500 HPL (Rmax). Derived figures are our arithmetic on stated inputs, labeled [derived].

03 · THE STACK, DRAWN TO SCALE

Seven layers of savings.
One crack through all of them.



Each layer is published and citable — the strongest verified number is drawn on its stone. And each layer's own literature names the same hole: the trigger, the variance, the committee estimates *where the model is probably wrong*. None can say *where physics is definitely violated*.

One strongest number per layer, each traced in its digest (page 15). AdsorbML (Lan et al., npj Comput. Mater. 2023); Jinnouchi/Kresse PRB 100, 014105; arXiv:2405.04967; Garrido Torres 2019 PRL 122, 156001 (5–25×); REST2 trpcage; DeepH MoS2 supercell; ACE/ACE-ISDF 1,000-atom Si HSE.

04 · SEVEN LAYERS, ONE HOLE

What each layer saves — and what each one cannot say.

01	**Surrogate transition-state search** 5–25× fewer calls, cost decoupled from image count · Garrido Torres 2019; Koistinen/Jónsson 2017; ~10× on 500 molecular reactions (Goswami 2025)	**THE HOLE** GP variance is a sampling-density signal, not an accuracy bound. Each search re-trains its surrogate, then discards it.
02	**Active learning** >99% of first-principles steps skipped (VASP MLFF) · DP-GEN labeled 0.0044% of ~650M configurations · FLARE: ~100–250 calls	**THE HOLE** Every mainstream trigger is sufficient-but-not-necessary — DP-GEN's own authors concede the committee can agree and be wrong. Rare events are the documented miss.
03	**Δ-learning, multi-fidelity, transfer** MatterSim revPBE0-D3 water: 30 high-fidelity configurations vs 900 from scratch · DPA-2: 1–2 orders less downstream data	**THE HOLE** Fine-tuning reintroduces PES softening, fails out-of-distribution, and costs a training run per model per chemistry — the wrong shape for breadth.
04	**Abstention economics** AdsorbML: DFT-level adsorption minimum in 87.4% of ~1000 systems, full DFT only on the ~13% tail	**THE HOLE** No published budget prices abstention across models — every number is per-model. Page 07 gives our projected count for one panel.
05	**Systems-level DFT acceleration** GPU ports 3–20× per node · reduced-rank exact exchange ~2 orders · Periodic Pulay 3× fewer SCF iterations · ML density guesses −20% total cost	**THE HOLE** Cuts the cost per evaluation, not the count — and the aggressive settings carry exactly the silent-failure risk that gates exist to catch.
06	**Electronic-structure surrogates** ML Hamiltonians eliminate 100% of SCF at inference (DeepH 10^3×; HamGNN: 4,284-atom Si in 36 s) · M-OFDFT: chemical accuracy at 27.4×	**THE HOLE** Predicted Hamiltonians do not give variationally reliable energies/forces for MD or NEB — and ML functionals fail silently (DM21).
07	**Path & sampling algorithms** Growing-string TS guess in ~20–90 gradient calls · ART nouveau: 50 vs 463 force evaluations · REST2: 10 replicas vs 48 · OPES ~10×	**THE HOLE** Cheap exploration distorts or skips the TS region, and string methods converge to second-order saddles at measured rates — a wrong-curvature failure a curvature check

05 · THE BIG NUMBER

72.4%

fewer *projected* DFT evaluations when four models share their anchors instead of each running its own.

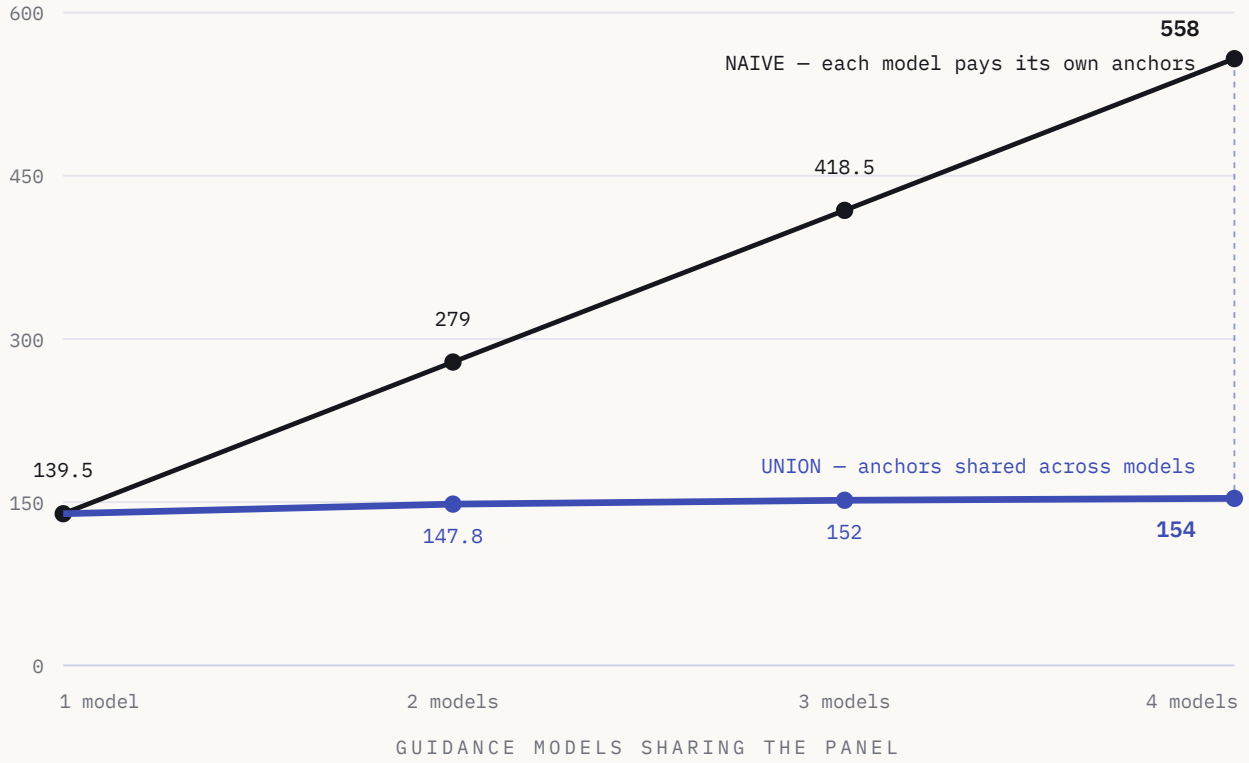
Projected accounting, no new DFT: 154 instead of 558 projected DFT evaluations on 29 analyzable diffusion paths of the locked Z1 panel — 404 fewer, a 3.62× reduction. The baseline is four separate runs. The paths have 5–7 images, so the union covers most images and deduplication alone caps the saving near 73%.

It is a count of evaluations, not an accuracy result. Page 10 reports what happened when we ran the pilot.

z1-union-anchor-economics.json · sha256 22c0ba51d0c798c2… · recomputed 2026-07-21 from the Round-4 float64 campaign artifacts under the frozen anchor rule of the preregistered sparse-DFT pilot. 29 analyzable paths: 18 of 5 images and 11 of 7, 167 images in all; the union covers 154 of them, and identical anchors across all four models would give 150 (73.1% fewer).

06 · THE CHART

Add models. The projected bill barely moves.



Four models of guidance cost *~10% more projected DFT* than one.

Mean over model subsets as the ensemble grows from one to four models: the naive bill multiplies almost exactly 4× — 139.5, 279, 418.5, 558 — while the shared union grows 139.5, 147.8, 152, 154, and is already flat at three models. Models largely nominate the same images; each added model costs its full naive share but adds few *new* anchors. On paths this short (5–7 images), most of the saving is simply not evaluating the same image four times.

Union-anchor scaling, mean over subsets of the 29 analyzable Z1 paths; cumulative fixed order (chgnet, +mace-mp-small, +mace-mp-medium, +mace-mpa-0-medium) agrees: 142, 144, 152, 154. Record: z1-union-anchor-economics.json and docs/analysis note in the pack.

07 · THE PROJECTION, STATED PLAINLY

One panel, four models, and an honest denominator.

The locked Z1 panel holds 30 diffusion paths. Path 14 (`mp-756912_1_1_1_0_0`) failed CI-NEB convergence under the frozen protocol in *all four* model artifacts, so no profile exists to guide anchors — the analyzable denominator is 29 of 30, on record, with per-model failure records in the JSON. Three more paths carry partial coverage; agreement there is reported under two conventions. Nothing was imputed.

PROJECTED DFT EVALUATIONS UNDER THE FROZEN ANCHOR RULE — RECORDED ARTIFACTS ONLY, NO DFT RUN

BASIS	PATHS	NAIVE	UNION	SAVED	RATIO
All analyzable paths (primary)	29/30	558	**154**	**72.4%**	3.62×
Fully-covered paths only	26/30	520	136	73.8%	3.82×

DO THE MODELS EVEN AGREE? PREDICTED SADDLE/EXTREMA COINCIDENCE OVER 29 PATHS

STATISTIC	AVAILABLE-MODELS	ALL-FOUR-REQUIRED
Saddle (argmax) exact	23/29	20/29
Saddle within ±1 image	26/29	23/29
Both extrema exact	10/29	9/29
Both extrema within ±1	12/29	11/29

The anchor rule is frozen: per model and path, endpoints plus the model-predicted minimum and a window around the model-predicted maximum. The naive total pays every model's set; the union total pays one evaluation per unique image per path. Before any DFT ran, a simulation of the sparse protocol on the recorded reference profiles reproduced the reference barriers with an MAE of 1.2–9.4 meV at ~7 anchors per path. The DFT test came next; page 10 reports it.

DERIVED ESTIMATE — UNDEMONSTRATED

Union-sparse evaluation (154 anchors vs a 4-model dense accounting of 200–400 per path) is a ~38–76× reduction in evaluation count; per-anchor levers from the systems layer (mixed precision ~2×, ML density guess ~1.25×) compose orthogonally to ~95–190×. Each factor is individually cited; *the product has not been demonstrated.* The July 2026 pilot did not demonstrate it: on these 5–7-image paths it evaluated every image, and its preregistered gate failed.

Method, reconciliation of the retracted earlier figures, and per-path tables: z1-union-anchor-economics.md in the pack. Anchor rule mirrored byte-exact from the frozen pilot implementation.

08 · WHAT HAPPENED WHEN WE RAN IT

The July 2026 pilot. Its preregistered gate failed.

In July 2026 we ran the sparse-anchor pilot with GPAW on the same 30-path panel. These are realized results on a different set of paths from the projection — 23 paths ran and seven large-cell paths were deferred — and the two are not combined.

THE PILOT'S CHECKS, IN THE ORDER THEY WERE SET

CHECK	BASIS	RESULT
Preregistered gate, 2026-07-20	Sparse GPAW barrier vs the panel's published VASP reference; 40 meV gate	**Failed: 118.8 meV on path 7**
Amended gate, 2026-07-21	Sparse vs dense GPAW profile of the same images; written after the path-7 failure	0.0–6.8 meV, all four models; same-engine only
VASP line, non-gating	Mean absolute error against the published references	**693.0–705.7 meV**
Large-cell paths	Seven paths deferred	No verdict

The gate was moved after the failure. Amendment 01, dated 2026-07-21 and written after the path-7 result, made the primary gate a same-engine comparison; the VASP comparison stayed as a secondary, non-gating line. Amendment 02, the same day, loosened two GPAW settings after a one-path revalidation. Because amendment 01 came after an outcome was seen, everything that passed under it is outcome-dependent.

The same-engine pass is a self-consistency check, not an accuracy result. On the 23 paths that ran, every image was evaluated, because these paths have 5–7 images and the amendment extended the anchor set to the full path. With every image evaluated, agreement with the dense same-engine profile is close to guaranteed by construction. We count it as Open (same-engine only).

Against the published references the pilot was about 700 meV off: a mean absolute error of 693.0 meV for CHGNet and MACE-MP-0 medium and 705.7 meV for MACE-MP-0 small and MACE-MPA-0 medium, larger than the raw models' own Round-4 barrier errors. The GPAW–VASP energy offset varied from image to image on 22 of the 23 paths, so on this panel the two engines' barriers cannot be compared directly.

> WHAT THIS LEAVES
>
> The 72.4% is an accounting result on short paths, and the one realized test of sparse anchors did not show accuracy against published references. A fair test needs longer paths and a reference the engine can match.

Sources (lupine-rhizo): data/candidates/z1-union-campaign.json, recorded 2026-07-24; docs/plans/2026-07-20-sparse-dft-pilot-preregistration.md; amendments 01 and 02, docs/plans/2026-07-21-sparse-dft-pilot-amendment-0{1,2}.md. Full account: lupine.science/articles/shared-dft-anchors/.

09 · DIRECTION: HOW A COMMONS COULD COMPOUND

“Every team that joins makes every other team faster.”
A direction, not a result.

A physical-law check — the curvature signature at a first-order saddle, energy–force consistency, symmetry constraints, boundary conditions — is a non-rival good: contributed once, it could gate other teams’ simulations too, at zero marginal sharing cost. Unlike training data, *such a check leaks nothing about the contributor’s chemistry or commercial targets*. Every failed simulation is a candidate check, and formalizing your own failure could turn your most expensive knowledge into an asset for a commons that pays you back in everyone else’s checks.

1. **Run your campaign against shared checks.** Failure modes that existing checks already cover would be caught, saving evaluations.
2. **Your novel failures get formalized.** Each new failure mode becomes a new contributed check.
3. **Every member’s gates get sharper.** False-accept and false-reject rates would fall for everyone.
4. **Savings compound.** If gates sharpen as members join, evaluations saved grow with membership.

A commons would have two tiers, and members would choose their exposure. **Checks** are zero-leak by construction. **Anchor libraries** — a content-addressed store mapping structure-hash to DFT energy and forces — would be opt-in per project: one team’s anchor could serve every other team’s model on the same path, under the arithmetic projected on page 07.

Direction: this page describes where we want the work to go; it is not a result. The projected union-anchor count within one lab (72–74%, sub-linear in model count) is the evaluation-side version of the idea; the July 2026 pilot did not show that sparse anchors are accurate (page 10). Lean 4 checks the arithmetic and stated inequalities behind published numbers; it does not check the physics, the reference data or the pipeline.

10 · THE DEMAND SIDE ALREADY EXISTS

Seven digests, working independently, came back with the same hole. Read the literature's own concessions:

Learned triggers are *sufficient-not-necessary* — the committee can agree and be wrong (DP-GEN). GP variance *underestimates* error under strong extrapolation (FLARE). Ensemble σ *decorrelates* from true error. Fast exploration *skips* the transition-state region (OneOPES). String methods land on *second-order* saddles at measured rates. ML functionals fail *silently* (DM21).

Our hypothesis is that many of those failure modes are *physical-law violations a check could detect without learning anything*. If so, a commons of such checks would meet a demand the field's own literature keeps describing.

11 · RESTRAINT, AS A POLICY

What we do not claim.

01 Not accuracy.

The 72.4% counts evaluations; it says nothing about whether sparse anchors give accurate barriers. The pilot that tested this failed its preregistered gate and was about 700 meV off the published references (page 10).

02 Not generality beyond the basis.

The projection rests on one 30-path barrier panel, four universal MLIPs and one chemistry family; the pilot used one DFT engine (GPAW). The union scaling curve is four points deep.

03 Not the stacking figure as a measurement.

The ~38–76× (and composed ~95–190×) evaluation-count reductions are derived estimates — our arithmetic on individually cited factors — labeled derived wherever they appear. The July 2026 pilot did not demonstrate them.

04 Not victory over the point solutions.

For a single isolated saddle search with a cheap oracle, local GP surrogates win outright. For one stable, high-volume chemistry, fine-tuning is the right tool. Our wedge is amortized, multi-path, multi-model workloads — panels, networks, screening campaigns.

05 Not blanket validation of every literature number.

The digests retain their [UNVERIFIED] claim flags and per-source access levels. The audit verified every identifier — it does not upgrade abstract-only evidence or independently reproduce reported savings factors.

06 Not peer review.

This is a research release with committed, hashable records — an invitation to inspect and recompute, not a claim of external validation.

Restraint is the policy: a measurement that overclaims stops being one.

12 · THE CITATION AUDIT

Every identifier, checked against its registry.

All 163 unique identifiers extracted from the nine digests were verified against the official registries — arXiv IDs against the arXiv Atom API, DOI strings against Crossref Works metadata — with returned titles reviewed against the citing rows. That review caught one live but unrelated DOI hidden in a link target.

115/115	48/48	0
arXiv identifiers verified	DOI strings verified through Crossref	unresolved identifiers

Corrections applied on the record. The Chen et al. multi-fidelity citation was moved from a nonexistent DOI to Nature Computational Science 1, 46–53 (10.1038/s43588-020-00002-x). The Nandi 2021 link target — which pointed at a live but unrelated paper — was rebound to the citation text's own DOI (10.1063/5.0038301). And the VASP MLFF venue was reconciled: PRB 100, 014105 / arXiv:1904.12961 is the melting-point paper behind the >99%-bypass claim; PRL 122, 225701 is the distinct earlier hybrid-perovskite demonstration.

> **WHAT THE AUDIT DOES NOT DO**
>
> Identifier verification does not upgrade abstract-only evidence, does not remove each digest's [UNVERIFIED] quantitative-claim flags, and does not independently reproduce reported savings factors. It says the citations resolve to what we say they are — no more, and no less.

Full audit record: citation-verification-2026-07-21.md in the pack, with the per-identifier machine-readable artifact and the reproducible checker named there.

13 · DATA & REPRODUCTION

Download it. Hash it. Recompute it.

The Savings Stack v1 pack ships the ten chapters, the machine-readable record, the analysis script, and the audit — with a sha256 manifest over every file. Since 2026-11-11 our first-party text, figures and data are licensed CC BY 4.0 and our code MIT; cited literature keeps its own terms. The pack's README, released 2026-07-21, still states ODbL-1.0, because the pack is hash-locked.

savings-techniques-synthesis-2026-07-21.md	the synthesis chapter
savings-surrogate-neb.md	digest · layer 1
savings-active-learning.md	digest · layer 2
savings-delta-multifidelity.md	digest · layer 3
savings-abstention-economics.md	digest · layer 4
savings-dft-systems.md	digest · layer 5
savings-electronic-surrogates.md	digest · layer 6
savings-path-sampling-algorithms.md	digest · layer 7
supercomputing-atomistic-history.md	backdrop · the cost wall
research-compute-resourcing.md	backdrop · access & resourcing
z1-union-anchor-economics.json (+ .sha256)	the machine-readable record
z1-union-anchor-economics.md	method & reconciliation note
union_anchor_economics.py	the reproducible analysis
citation-verification-2026-07-21.md	the identifier audit
MANIFEST.sha256	hashes over every file

```
$ sha256sum -c MANIFEST.sha256          # every file: OK
$ python3 union_anchor_economics.py --local <artifacts> \
    --out z1-union-anchor-economics.json   # recompute the record
```

Pack: lupine.science/data/savings-stack-v1/ · Booklet: lupine.science/booklets/the-savings-stack.pdf · Full scientific detail: library.lupine.science, group "savings-stack".



Evidence before claim.

The Savings Stack · v1.1

Booklet & data pack — lupine.science

Full detail — library.lupine.science · group “savings-stack”

The instrument — lupi.live